23. It’s interesting to note that, years before the supernova results, prescient theoretical works by Jim Peebles at Princeton, and also by Lawrence Krauss of Case Western and Michael Turner of the University of Chicago, and Gary Steigman of Ohio State, had suggested that the universe might have a small nonzero cosmological constant. At the time, most physicists did not take this suggestion too seriously, but now, with the supernova data, the attitude has changed significantly. Also note that earlier in the chapter we saw that the outward push of a cosmological constant can be mimicked by a Higgs field that, like the frog on the plateau, is perched above its minimum energy configuration. So, while a cosmological constant fits the data well, a more precise statement is that the supernova researchers concluded that space must be filled with something *like* a cosmological constant that generates an outward push. (There are ways in which a Higgs field can be made to generate a long-lasting outward push, as opposed to the brief outward burst in the early moments of inflationary cosmology. We will discuss this in Chapter 14, when we consider the question of whether the data do indeed require a cosmological constant, or whether some other entity with similar gravitational consequences can fit the bill.) Researchers often use the term “dark energy” as a catchall phrase for an ingredient in the universe that is invisible to the eye but causes every region of space to push, rather than pull, on every other.

24. Dark energy is the most widely accepted explanation for the observed accelerated expansion, but other theories have been put forward. For instance, some have suggested that the data can be explained if the force of gravity deviates from the usual strength predicted by Newtonian and Einsteinian physics when the distance scales involved are extremely large—of cosmological size. Others are not yet convinced that the data show cosmic acceleration, and are waiting for more precise measurements to be carried out. It is important to bear these alternative ideas in mind, especially should future observations yield results that strain the current explanations. But currently, there is widespread consensus that the theoretical explanations described in the main text are the most convincing.

Chapter 11